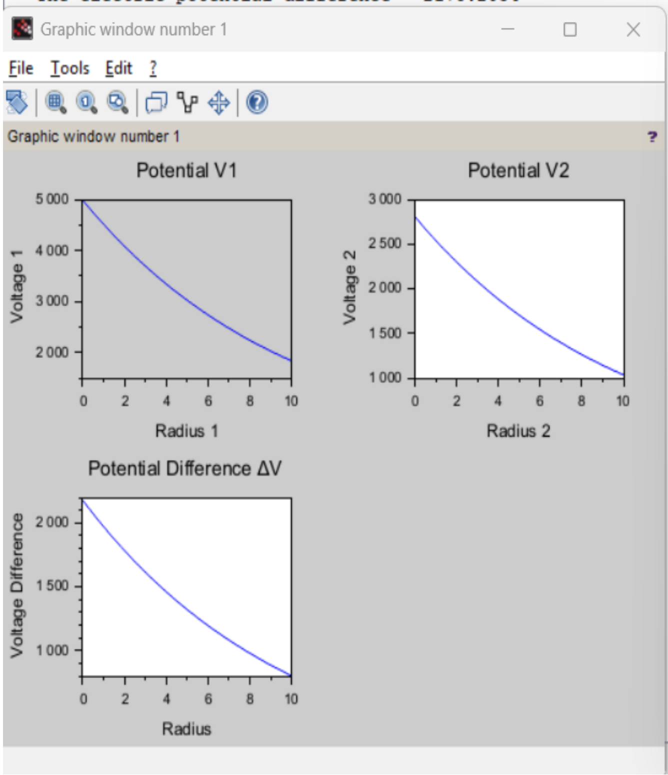
Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

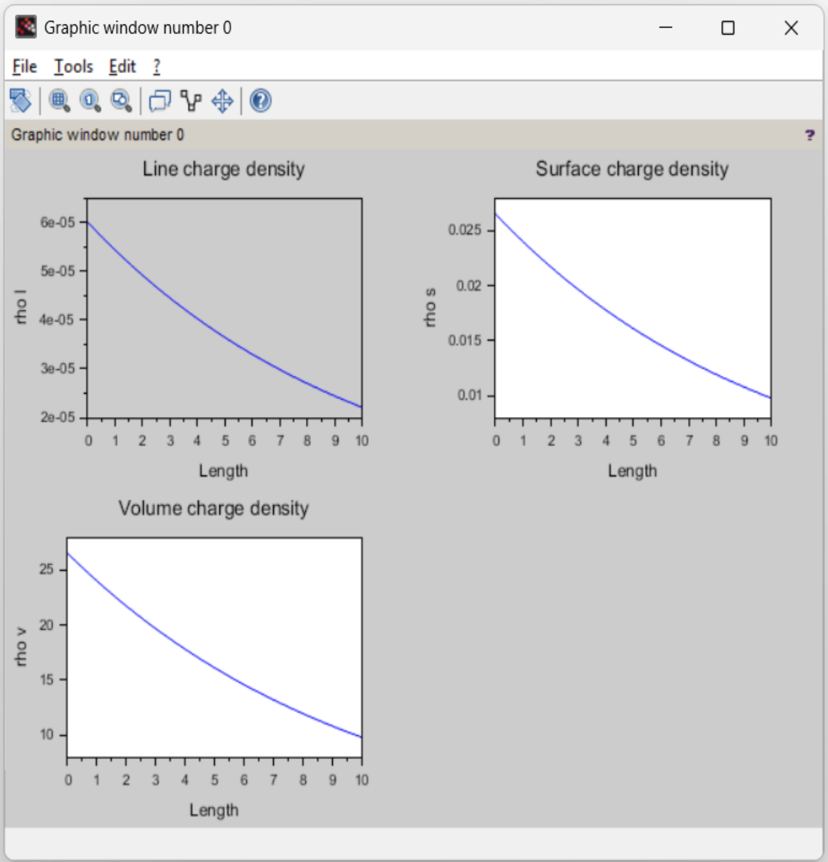
Program:

```
clc;
clear;
epsilon0 = 8.854e-12;
pi = %pi;   // use Scilab's built-in pi
// Input
q  = input("Enter the q value: ");
ax = input("Enter the value of ax= ");
ay = input("Enter the value of ay= ");
bx = input("Enter the value of bx= ");
by = input("Enter the value of by= ");
// Distances
r1 = sqrt(ax^2 + ay^2);
r2 = sqrt(bx^2 + by^2);
// Potentials
v1 = q / (4 * pi * epsilon0 * r1);
v2 = q / (4 * pi * epsilon0 * r2);
// Potential difference
dv = v1 - v2;
// Output
disp("The value of v1 = " + string(v1));
disp("The value of v2 = " + string(v2));
disp("The electric potential difference = " + string(dv));
// Plotting
r = linspace(0, 10, 50);
y = v1 * exp(-0.1 * r);
z = v2 * exp(-0.1 * r);
x = dv * exp(-0.1 * r);
figure();
subplot(2,2,1);
plot(r, y);
xlabel("Radius 1");
ylabel("Voltage 1");
title("Potential V1");
```

Output:

```
Enter the q value: 2*10^-6

Enter the value of ax= 2

Enter the value of ay= 3

Enter the value of bx= 4

Enter the value of by= 5

  "The value of v1 = 4985.5025"
  "The value of v2 = 2807.2991"
  "The electric potential difference = 2178.2034"
```





Calculation:

Exp:-5

Enter q value :- 2×10^{-6}

Enter value of ax:- 2

Enter value of ay:- 3

Enter value of bx:- 4

Enter value of by:- 5

A) From the given values

$\varepsilon_0 = 8.854 \times 10^{-12}$

$\pi = 3.14$

i) Distance

$$r_1 = \sqrt{(ax)^2 + (ay)^2} = \sqrt{2^2 + 3^2} = \sqrt{13} = 3.606$$

$$r_2 = \sqrt{(bx)^2 + (by)^2} = \sqrt{4^2 + 5^2} = \sqrt{41} = 6.403$$

ii) Potentials:-

$$V_1 = \frac{q}{4\pi \varepsilon_0 r_1} = \frac{2 \times 10^{-6}}{(4 \times 3.14 \times 8.854 \times 10^{-12})(3.606)} = 498\cancel{8}5.502$$

$$V_2 = \frac{q}{4\pi \varepsilon_0 r_2} = \frac{2 \times 10^{-6}}{(4 \times 3.14 \times 8.854 \times 10^{-12})(6.403)} = 2807.299$$

iii) Potential difference:-

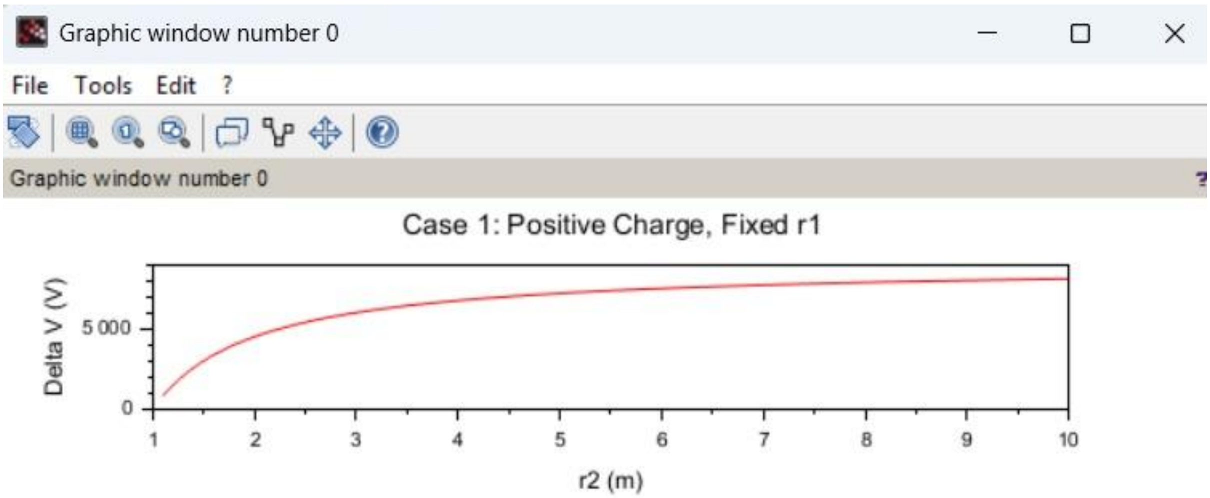
$$\Delta V = V_1 - V_2 = 4985.502 - 2807.299$$

$$\simeq 2178.2034$$

Case 1: A positive charge with fixed r1=1 m r2 varying.

```
// Constants
epsilon0 = 8.854e-12; // Permittivity of free space
k = 1 / (4 * %pi * epsilon0); // Coulomb's constant
q = 1e-6; // Charge in Coulombs (1 µC)

// Case 1: Positive charge, r1 = 1 m, r2 varies
r1_case1 = 1; // meters
r2_case1 = linspace(1.1, 10, 100); // meters
deltaV_case1 = k * q * (1 / r1_case1 - 1 ./ r2_case1);



// Plotting
clf();
subplot(3, 1, 1);
plot(r2_case1, deltaV_case1, "r");
xlabel("r2 (m)");
ylabel("Delta V (V)");
title("Case 1: Positive Charge, Fixed r1");

```

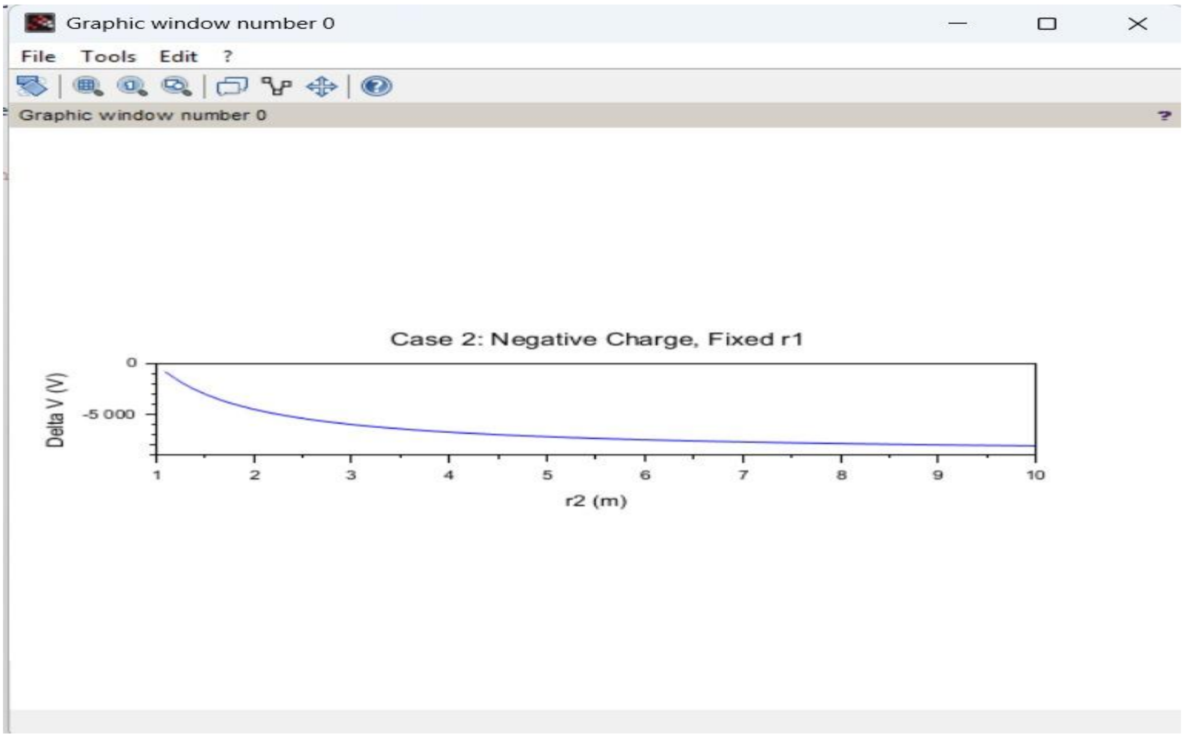
Output:



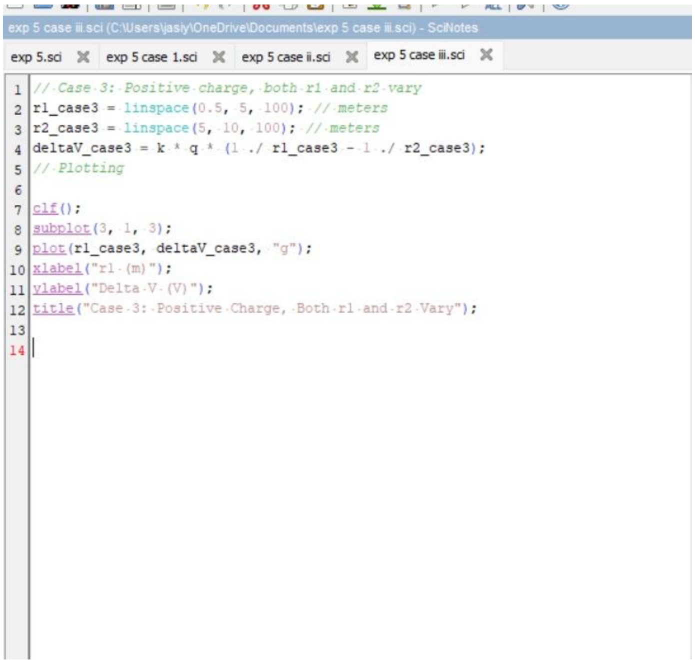
Case 2: A negative charge with r1r_1r1 fixed and r2r_2r2 varying.

```
// Case 2: Negative charge, r1 fixed, r2 varies
q_case2 = -1e-6; // -1 µC
r1_case2 = 1; // meters
r2_case2 = linspace(1.1, 10, 100); // meters
deltaV_case2 = k * q_case2 * (1 / r1_case2 - 1 ./ r2_case2);
// Plotting

clf();
subplot(3, 1, 2);
plot(r2_case2, deltaV_case2, "b");
xlabel("r2 (m)");
ylabel("Delta V (V)");
title("Case 2: Negative Charge, Fixed r1");
```

Output:



Case 3: Both points r1r_1r1 and r2r_2r2 vary with a positive charge.




```
// Case 3: Positive charge, both r1 and r2 vary
r1_case3 = linspace(0.5, 5, 100); // meters
r2_case3 = linspace(5, 10, 100); // meters
deltaV_case3 = k * q * (1 ./ r1_case3 - 1 ./ r2_case3);
// Plotting

clf();
subplot(3, 1, 3);
plot(r1_case3, deltaV_case3, "g");
xlabel("r1 (m)");
ylabel("Delta V (V)");
title("Case 3: Positive Charge, Both r1 and r2 Vary");

```

Output: